

Akhil Indraganti

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EDUCATION

University of North Texas Denton, Texas	2023 – 2025
Master of Science in Information Science Information Science	
VIT-AP University Amaravati	2018 – 2022
Bachelor of Technology in Electronics And Communication Engineering Electronics And Communication Engineering	

EXPERIENCE

Research Assistant, Full-Stack AI Developer University of North Texas Remote	June 2025 – Present
- Developed React/JavaScript frontend with real-time text selection and multi-round Q&A dialogs for scientific and financial reports, enhancing user interaction and data accessibility. - Engineered Node.js/ Express backend with OCR integration and RESTful APIs, managing MongoDB/ PostgreSQL databases and community-driven ranking algorithms for efficient data retrieval. - Integrated multiple LLM APIs with advanced prompt engineering and RAG functionality, improving information retrieval efficiency using Python and R for analysis.	
Descriptive Metadata Writer University of North Texas Denton, TX	Jul 2024 – May 2025
- Crafted metadata descriptions for over 1,500 digitized public records adhering to Dublin Core standards, ensuring information system consistency and discoverability. - Digitized cultural heritage materials using FADGI compliant scanners, contributing to accessible archival quality of historical documents. - Performed quality control reviews of digital image metadata and file organization, maintaining accuracy and integrity across collections through meticulous data validation.	
Software Engineer - I Visteon Bangalore, India	Jul 2022 – Jul 2023
- Enhanced system stability by identifying and resolving 10,000+ defects via static code analysis in C++ , improving software reliability and IT infrastructure performance. - Automated diagnostics and system tasks using Shell scripts across QNX/Linux environments, reducing manual overhead by 25%. - Collaborated in Agile teams using C++ to deliver features for automotive- grade digital instrument clusters, optimizing IT infrastructure and system performance.	
Internship Visteon Remote	Jan 2022 – Jun 2022
- Built a custom Linux distribution using Yocto and BitBake to enable Flutter UI on Raspberry Pi, reducing embedded platform prototyping time by 40%. - Developed and optimized build scripts to integrate Flutter SDK, streamlining deployment for embedded automotive systems. - Deployed functional Flutter applications with DRM backend, accelerating UI validation for a major automotive digital cockpit program.	

PROJECTS

Metadata Quality Evaluation and Improvement for Digital Archives using LLMs (Python, Tesseract OCR, LLaMA-4)	Jan 2025 – May 2025
- Developed a pipeline for automated metadata generation and enrichment of over 400 historical documents using Python and LLMs, improving information system accuracy and leveraging R for quality analysis. - Implemented few-shot prompting to enhance archival records, achieving a semantic similarity score of 0.83 against human-curated metadata. - Evaluated AI-generated metadata using precision, recall, and F1-score, demonstrating over 90% accuracy for structured fields and a 12% improvement in data consistency.	
Public Transport Efficiency Analysis (GCP, Hadoop, Spark, BigQuery, Python, R)	Jan 2025 – May 2025
- Architected a data analysis pipeline on GCP to assess Denton County Transportation Authority data, identifying inefficiencies in IT infrastructure and leveraging R for statistical analysis. - Utilized OpenRefine for data cleansing, Hadoop and Spark for scalable processing, and BigQuery for warehousing and SQL-based insights. - Benchmarked Spark as 4x faster than Hive for querying; visualized findings using R and Python and delivered actionable recommendations, showcasing effective query writing and data interpretation.	
Restaurant Database Management System (MySQL, SQL)	Aug 2023 – Dec 2023
- Designed a MySQL relational database modeling customers, employees, orders, and transactions, creating an ERD to enforce data integrity and efficient data management. - Implemented role-based access and staff scheduling to streamline operations, enhancing customer service through efficient information system design. - Authored complex SQL queries for business reporting such as daily revenue and reservation frequency, demonstrating strong query writing skills and data analysis proficiency.	
Speech Enhancement Using Deep Learning (MATLAB, Python, Librosa)	Aug 2021 – Dec 2021
- Developed source separation models to isolate vocal and instrumental tracks from mixed signals, enhancing audio data processing capabilities. - Applied FFT and STFT techniques along with soft masking to improve Music Information Retrieval (MIR) tasks. - Generated clean .wav outputs and spectrograms, enabling applications like karaoke track creation and audio analysis, showcasing	

technical computer science principles.

SKILLS

- **Programming Languages:** Proficient in Python and R for data analysis and IT infrastructure development, complemented by strong SQL for database management and C++ for systems programming.
- **Data Analysis & BI Tools:** Expertise in Tableau and Power BI for creating insightful business intelligence dashboards and leveraging statistical analysis techniques.
- **AI & Machine Learning:** Skilled in LLMs, RAG, and prompt engineering for extracting complex information, with experience in developing deep learning models.
- **Cloud & Big Data Platforms:** Experienced with GCP, Spark, and Hadoop for scalable data processing and warehousing, enhancing IT operations and data accessibility.